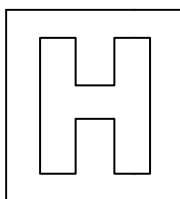


Candidate Name: _____

Class Adm No

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2014 Promotional Examination II Pre-university 2

H1 Biology

Suggested Answers

8875/02

17 September 2014

2 hours

Additional Materials: Writing papers

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, Adm No. and class on all the papers you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions on the answer booklet

Section B

Answer **one** question on the writing papers.

At the end of the examination, fasten all your work securely together. The number of marks is given in brackets[] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
Section B	
Total	

This question paper consists of 16 printed pages including 1 blank page

[Turn over

Section A

Answer **all** questions in this section.

1. Lactate dehydrogenase is an enzyme that is found extensively in body tissues such as the blood cells and muscle cells. Fig. 1.1 shows the structure of human lactate dehydrogenase.

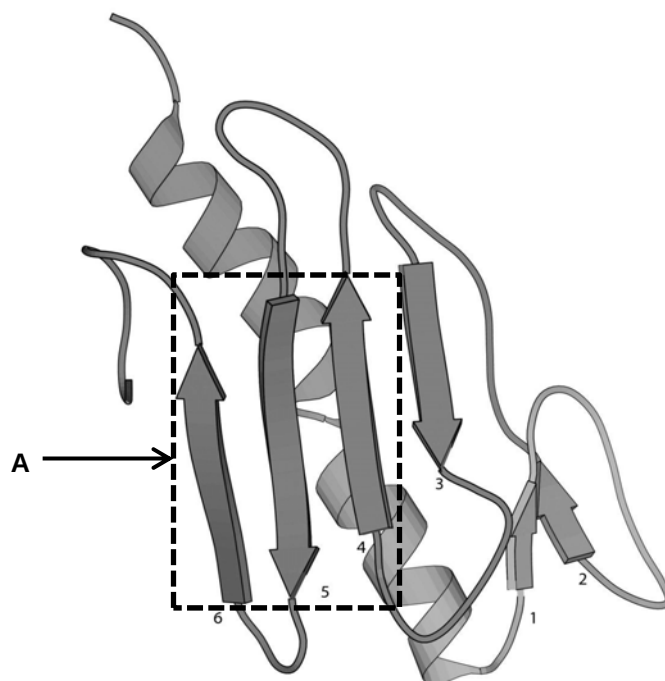


Fig. 1.1

- (a) (i) Identify structure A.

β -pleated sheet;

; for 1 mark, 1 mark max

- (ii) Describe how structure A is formed.

is made up of different sections of the polypeptide chain that run alongside each other/ are antiparallel;

Adjacent strands are held together by hydrogen bonds formed between CO and -NH groups in the polypeptide backbone to form the sheet;

; for 1 mark, 1 mark max

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Lactate dehydrogenase works by binding to its substrate pyruvate, which is a product of glycolysis. Lactate dehydrogenase catalyses the conversion of pyruvate into lactate. The toxic effects of heavy metals on organisms stems from their effect on metabolically-important enzymes, such as lactate dehydrogenase. Fig. 1.2 shows the effect of varying concentrations of mercury ions on the rate of reaction catalysed by lactate dehydrogenase isolated from a species of fish.

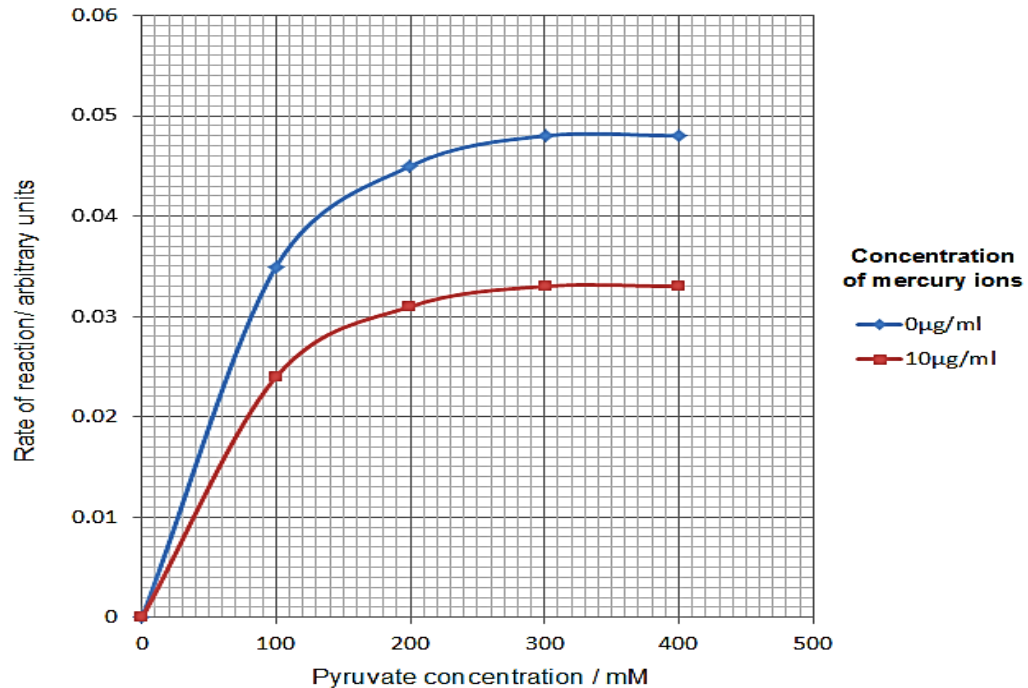


Fig. 1.2

(b) (i) Briefly describe the mode of action of lactate dehydrogenase.

1. Enzyme lowers activation energy;
 2. Ref to acting as orientation surfaces/ ref to strain effect/ proximity effect/ microenvironment effect/ acid-base catalysis;
 3. The (3D conformation of) substrate is complementary to the active site of the enzyme, substrate bind to enzyme to form enzyme-substrate complexes;
 4. With ref. to change in 3D conformation of enzyme for a better fit/ tighter binding between substrate and enzyme/ with ref to induced fit hypothesis
- ; for 1 mark, 3 marks max

- (ii) With reference to Fig. 1.2, describe and explain the effect of mercury ions on the rate of reaction catalysed by lactate dehydrogenase.

- In the presence of 10µg/ml of mercury, the maximum rate (V_{max}) of reaction decreased from 0.048 arbitrary units to 0.033 arbitrary units;
 - Mercury ions act as a non-competitive inhibitor;
 - Has a different structure from the substrate pyruvate and binds permanently / irreversibly to another site away from the active site on the enzyme;
 - Forms a strong covalent bond to an amino acid R group (e.g. -SH group) on the enzyme;
 - This causes a conformational change on the enzyme which alters the shape of the active site;
 - Hence the enzyme is rendered non-functional / no longer able to bind to substrate / decreased rate of formation of enzyme-substrate complex and hence decreasing the rate of reaction;
 - Increasing substrate/pyruvate concentration will not allow the maximum rate of reaction to be achieved;
- ; for 1 mark, 3 marks max

In nature, collagen is found exclusively in animals, especially in the flesh and connective tissues of mammals. Collagen is a part of the connective tissues in the skin that helps in maintaining skin firmness and suppleness. Collagen is a fibrous protein and has a very distinct structure and function from lactate dehydrogenase.

- (c) Contrast between the structure of collagen and lactate dehydrogenase.

- Amino acid sequence in collagen is highly regular with repeating tripeptide sequence while amino acid sequence in cellulase rarely shows regularity;
- Amino acid sequence in collagen may vary between two samples of collagen; while amino acid sequence in cellulase is highly specific;
- Length of polypeptide chain varies in collagen while length of polypeptide chain seldom varies in cellulase;
- More amino acids make up collagen while less amino acids make up cellulase;
- Modified amino acids such as hydroxylysine and hydroxyproline are found in collagen but not cellulase;

; for 1 mark, 2 marks max

[Total: 9]

2. Fig. 2.1 is an electron micrograph of chromatin found in the nuclei of eukaryotes. Most of the chromatin is seen to be in the form of a fiber with a diameter of about 30 nm.



Fig. 2.1

- (a) Describe how DNA is packed into the 30 nm chromatin fibre.

- The DNA is wrapped around groups of 8 histones to form nucleosomes;
- The DNA is negatively charged whereas the histone proteins are positively charged and are held together by ionic bonds;
- The remainder of the DNA, called the linker, joins adjacent nucleosomes;
- The nucleosomes and linkers constitute the chromatin fibre;
- The chromatin fibre coils around itself to form a solenoid, which is 30 nm chromatin fibre;

; for 1 mark, 2 marks max

In the 1930s, Hermann Muller was the first researcher to note that the ends of chromosomes had unique properties. Muller named these ends telomeres (from the Greek words *telo*, meaning "end," and *mere*, meaning "part"), based on their position on chromosomes.

There is a limit to the number of times a normal human cell population will divide. Empirical evidence shows that telomeres get slightly shorter with each round of DNA replication until they shorten to a critical length to trigger cell death.

Fig. 2.2 shows the position of telomeres in a DNA double helix.

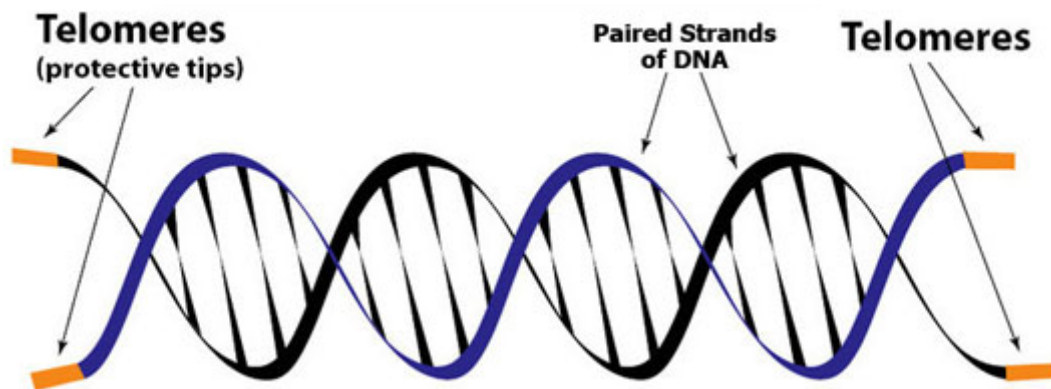


Fig. 2.2

(b) Explain why telomeres shorten with each round of DNA replication.

- Telomeres shorten in part because of the **end replication problem** that is exhibited during DNA replication;
- DNA polymerases move in the 5' to 3' direction, one finds a leading and a lagging strand on the DNA molecule being replicated;
- Eventually when the last primer in the lagging strand is removed by DNA polymerase I and replacing it with the correct DNA nucleotides;
- In the case at the position of the last primer, the 3'OH group is absent;
- RNA primer is removed but the gap is not replaced by DNA nucleotides;

; for 1 mark, 2 marks max

(c) State one role of telomeres in cells.

- The telomere regions delay the erosion of genes near the ends of chromosomes by allowing for the shortening of chromosome ends, which necessarily occurs during replication;
- Telomere shortening in humans can induce senescence, which blocks cell division;
- This mechanism appears to prevent genomic instability and development of cancer in human aged cells by limiting the number of cell divisions;

; for 1 mark, 1 mark max

Fig. 2.3 and Fig. 2.4 show the process of DNA replication and transcription in eukaryotes.

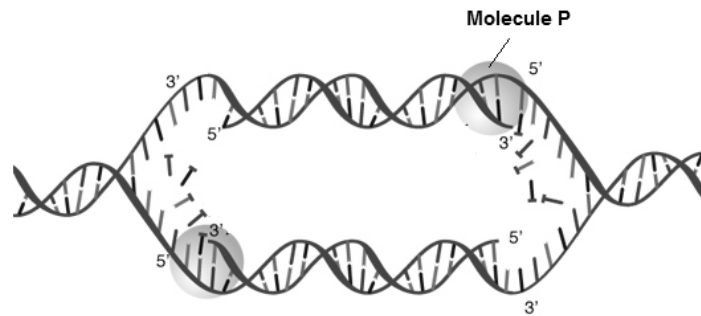


Fig. 2.3

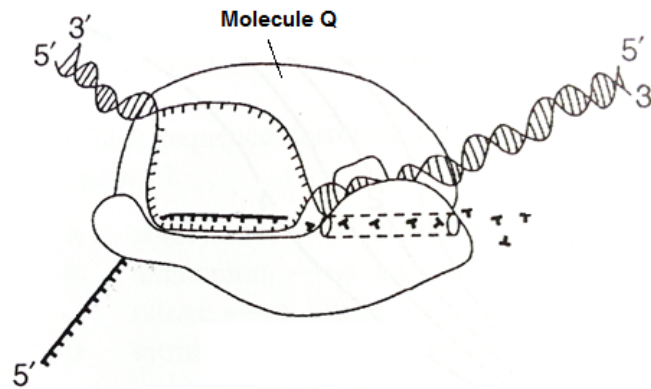


Fig. 2.4

- (d) Explain why RNA primer is required for the synthesis of new strand in DNA replication.

- DNA polymerase can only add nucleotides onto existing chain with free 3' OH group at one end;

; for 1 mark, 1 mark max

- (e) With reference to Fig. 2.3 and Fig. 2.4, describe **two** ways in which the functions of molecules **P** and **Q** are similar to each other.

Type of bond formed: phosphodiester bond;
 Template used: DNA as template;
 Direction of reading the template: from 3' to 5' direction;
 The product are formed in 5' to 3' direction;

; for 1 mark, 2 marks max

[Total: 8]

3. The term 'Manx syndrome' is used to describe a range of conditions seen cats with a shortened or absent tail. This syndrome arises as a result of a gene mutation in Manx gene. Manx gene determines the development of the cat's spine. Unfortunately, this mutation is often linked to some severe developmental abnormalities in the spinal cord leading to significant neurological disease.

Fig. 3.1 shows the sequence of bases in a section of Manx gene coding for the first seven amino acids of Manx polypeptide chain.

5' AAAGTGTCTGCTGGAACAT 3'

Fig. 3.1

- (a) (i) Using the space below, list the corresponding codons in the mRNA transcribed from the template in Fig. 3.1.

5' AUG UUC CAG CAG AAA CAG UUU 3'
; for 1 mark, 1 mark max

[1]

- (ii) Explain why a single base deletion in the DNA sequence is more likely to produce non-functional proteins than a single base substitution.

- Single base deletion leads to frameshift mutation;
- The entire reading frame for the nucleotide sequence after the point of mutation alters;
- Results in different codons coding for different amino acids, amino acids sequence is changed entirely;
- Single base substitution results in missense mutation;
- Change of one codon, change in one amino acid;
- or may result in silent mutation that results in no change in aa (if it occurs at 3rd base as a result of degeneracy of genetic code) Or aa has similar property

; for 1 mark, 2 marks max

(iii) Describe how mutation in Manx gene results in the abnormal phenotype in cats with Manx syndrome.

- Mutation in the Manx gene might lead to missense and nonsense mutation;
- Change in amino acid sequence or early termination of the translation process leading to a truncated polypeptide;
- Amino acid sequence carries the information that determine the folding of polypeptide chain;
- Disrupt 3D conformation, loss of function of the Manx protein, leading to the abnormal development of spinal cord;

; for 1 mark, 2 marks max

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The allele S^N codes for normal spine which results in long tail being form and S^A codes for abnormal and short spine. Cats that are homozygous recessive will not have their spines developed and so they will die early in development. Manx cats are tailless and they have a heterozygous genotype. Fig. 3.2 shows a cat with Manx syndrome.



Fig. 3.2

A second gene which controls for the appearance of the coat is located on the X chromosome. Allele D codes for 'dappled' coat (D) while allele d codes for 'plain' coat (d). A male Manx cat with dappled coat was mated on several occasions to the same female and a large number of offspring consisted of the following phenotypes, as summarised in Table 3.1 is observed.

Table 3.1

Tail Type	Gender	Coat colour	% no. of kittens
Long tailed	Male	Dappled	12.5
Long tailed	Female	Dappled	12.5
Manx	Male	Dappled	25
Manx	Female	Dappled	25
Did not survive development			25

- (b) Using suitable symbols, construct a genetic diagram to explain the results in Table 3.1.

Let S^N be the (co)dominant allele that codes for normal spine, long tailed
 S^A be the (co)dominant allele that codes for abnormal, short spine
 X^D be the sex chromosome allele for dappled coat
 X^d be the sex chromosome allele for plain coat

Parental phenotypes : Male Manx cat, dappled coat x Female Manx cat, dappled coat

Parental genotypes (2n) : $S^N S^A X^{DY}$ x $S^N S^A X^D X^D$

F₁ gametes : $S^N X^D$ $S^N Y$ $S^A X^D$ $S^A Y$ x $S^N X^D$ $S^A X^D$

Punnett square to show random fusion of gametes by the F₁ generation:

♂ gametes	$S^N X^D$	$S^N Y$	$S^A X^D$	$S^A Y$
♀ gametes	$S^N S^N X^D X^D$ Female Long tailed Dappled coat	$S^N S^N X^D Y$ Male Long tailed Dappled coat	$S^N S^A X^D X^D$ Female Manx Dappled coat	$S^N S^A X^D Y$ Male Manx Dappled coat
	$S^N S^A X^D X^D$ Female Manx Dappled coat	$S^N S^A X^D Y$ Male Manx Dappled coat	$S^A S^A X^D X^D$ Dies	$S^A S^A X^D Y$ Dies

Offspring Genotypic Ratio	$S^N S^N X^D X^D$	$S^N S^N X^D Y$	$S^N S^A X^D X^D$	$S^N S^A X^D Y$	$S^A S^A X^D X^D$	$S^A S^A X^D Y$
	1	1	2	2	1	1
Offspring Phenotypic Ratio	Female Long tailed Dappled coat	Male Long tailed Dappled coat	Female Manx Dappled coat	Male Manx Dappled coat	Dies	Dies
Percentage evident in Table 2.1 (%)	12.5	12.5	25	25	12.5	12.5

[4]

[Total: 9]

4. Red fox and kit fox are two different species of fox that belong to the family of family Canidae, which are found commonly in North America.

The red fox, *Vulpes vulpes*, resembles a small dog with a long bushy tail, large pointed ears, a pointed muzzle and a red coat. On the other hand, the kit fox, *Vulpes macrotis*, is the smallest species of the family. It has large ears that help the fox lower its body temperature and give it exceptional hearing. Red fox can be found in the farmlands and forests of North America as compared to the kit fox, which lives on the plains and in deserts.

Through a series of studies, the red fox and kit fox are found to be evolved from a common ancestor.

- (a) Explain how natural selection could have caused the evolution of red fox and kit fox in North America.

Mutation

- Spontaneous mutation results in genetic variations among the fox in the ancestral population

Natural Selection

- Difference in selection pressure due to the different habitat that these forms are found;
- Red coat provide greater camouflage for the red fox in the farmland and forest while kit fox has large ears to lower body temperature in deserts where the temperature is high during the day;
- Red fox population has a selective advantage in forests and farmland while kit fox has a selective advantage to survive in desert;
- They will survive, reproduce and pass on their favourable genes to offsprings;
- Geographical isolation prevents interbreeding
- Over time, the two different population accumulate more genetic variations;

; for 1 mark, 3 marks max

Our understanding of the evolutionary relationship of species has been greatly improved through the use of molecular techniques. Many evolutionary studies use mitochondrial DNA rather than nuclear DNA.

(b) Explain why mitochondrial DNA is used to determine phylogeny instead of using nuclear DNA.

- Unlike nuclear DNA, whose genes are rearranged in the process of recombination, there is usually no change in mtDNA from parent to offspring.
- the mutation rate of mtDNA is higher than that of nuclear DNA and is easily measured and compared allows mtDNA to be a powerful tool for tracking the ancestry of many species back hundreds of generation;
- Common in all species; valid basis for comparison between distantly related species;

; for 1 mark, 2 marks max

(c) Explain how you could use molecular homology to show that red fox and kit fox evolved from a common ancestor.

- For basis of valid comparison the sequence of similar genes (for example, cytochrome c oxidase gene) are compared amongst the red and kit fox
- Or sequence of amino acids in the polypeptide of a named protein.
- Degree of similarity in sequences/ degree which sequences of gene are conserved between species are interpreted as DNA evidence for evolution.
- Generally, a higher degree which sequences are being conserved a higher degree of relatedness different species are to one another and vice versa.

; for 1 mark, 2 marks max

5. Dwarfism is genetic condition characterised by restricted growth and short stature. The average adult height among people with dwarfism is 122 cm. A significant proportion of dwarfism-related disorders are due to growth hormone deficiency. Growth hormone deficiency is treated with injections of a synthetic version of human growth hormone (hGH).

Fig 5.1 shows the pCMV6-XL5 plasmid. It is a plasmid with a multiple cloning site (MCS) that lies downstream of the CMV promoter. This plasmid can be inserted into both eukaryotic and prokaryotic host cells. The arrows denote the direction in which the genes are transcribed.

An artificially-synthesised hGH gene with flanking *Hind*III restriction site sequences was created. The restriction sites for the restriction enzymes *Hind*III and *Sac*I are shown in Fig. 5.2.

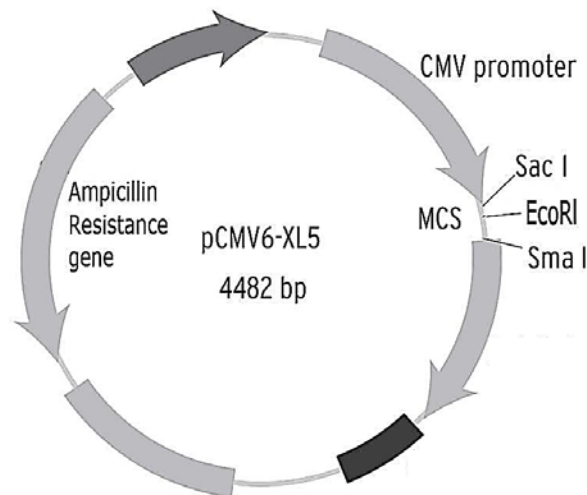
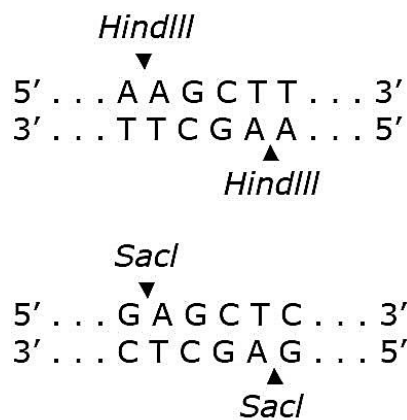


Fig. 5.1



▲ - point at which restriction enzymes cleave DNA.

Fig. 5.2

- (a) Using information in Fig. 5.1 and 5.2, explain how the hGH gene can be inserted into pCMV6-XL5.

- Cleave gene using HindIII to generate sticky ends;
- Cleave plasmid with SacI to generate complementary sticky ends to the HindIII sticky ends flanking the hGH gene;
- Mix the cleaved gene and plasmid together and add DNA ligase to seal the nicks / form phosphodiester bond between gene and plasmid;

; for 1 mark, 3 marks max

- (b) Outline the steps required to select bacterial colonies containing the human growth hormone gene after transformation has taken place.

- Spread/plate transformed bacteria onto agar/nutrient medium plates containing ampicillin;
- Bacterial colonies on the ampicillin plate are transferred onto a nitrocellulose membrane;
- Add sodium hydroxide to the filter to lyse bacterial cells and denature dsDNA into ssDNA;
- radioactively-labelled gene probe is added so that it will hybridize with the complementary DNA sequence of the human growth hormone gene;
- Wash away excess probe;
- The position of the cell containing the DNA-gene probe hybrid/labelled gene can be located by autoradiography performed on the filter;
- The position of dark spots on the x-ray film corresponds to the position of the bacterial colonies containing the human growth hormone gene;

Accept: Fluorescent-labelled gene probes and subsequent corresponding steps;

; for 1 mark, 3 marks max

- (c) State one problem of cloning human genes in bacteria.

- Presence of introns in human gene, cannot be spliced out in bacteria
- Lack of post-translational modification in bacteria, protein produced may not be functional

; for 1 mark, 1 mark max

[Total: 7]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc. as indicated in the question.

6.

- (a) Distinguish between the behaviours of homologous chromosomes in mitosis and the first meiotic division. [8]

STAGE	MITOSIS	FIRST DIVISION OF MEIOSIS
Prophase	No synapsis / no pairing of homologous chromosomes No crossing over / no chiasma formation	Synapsis occurs / homologous chromosomes pair up to form bivalents Crossing over occurs between non-sister chromatids of homologous pair at chiasmata.
Metaphase	Chromosomes arrange lengthwise / on the same plane / in a single row on the equator of spindle. No independent assortment since sister chromatids are identical.	Homologous chromosomes lie side-by-side / equidistant on either side of the equator. Independent assortment of alleles occur.
Anaphase	Centromere of each chromosome divide Sister chromatids separate OR Separating chromatids are identical	Centromeres do not divide Homologous chromosomes separate OR Separating chromosomes are not identical.

(b) With reference to specific genes, discuss how mutations can lead to cancer. [6]

- Gene mutations arise as a result of substitution, insertion, deletion and inversion;
- Substitution could result in missense mutation;
- Deletion and insertion of one nucleotide could lead to frame-shift mutation;
- Mutations in proto-oncogene and tumour suppressor gene result in uncontrolled division of cells;
- Proto-oncogenes promote the normal growth and division of cells;
- Proto-oncogenes such as Ras gene codes for Ras protein which is involved in promoting mitosis in the presence of signaling molecules such as epidermal growth factor;
- Gain-of-function mutations change the proto-oncogene to oncogene, causing production of hyperactive Ras protein or over-production of Ras protein.
- Tumor suppressor genes code for products that prevent or suppress cell growth and division;
- Such as p53 gene codes for p53 protein which is a transcription factor switching on genes whose products are involved in repairing DNA damage or triggering apoptosis should the damage is beyond repair;
- Point mutations and chromosomal mutations can lead to loss-of-function mutation in p53 gene, leading to nonfunctional p53 protein or inadequate amount of p53 protein;

; for 1 mark, 6 marks max

(c) Relate the structure of cell surface membrane to its properties. [6]

Structure	Property – Selective permeability
• comprises of phospholipid bilayers; • hydrophobic tails facing inwards and hydrophilic heads facing the aqueous medium; • hydrophobic barrier;	Permeable to non-polar / hydrophobic substances; [T o
• interspersed with proteins; • integral proteins are embedded in the membrane and peripheral proteins are not embedded in the membrane;	Transmembrane proteins/protein channels/protein carriers allow transport of ions/ polar/ hydrophilic substances across the membrane; : 2
Structure	Property – Fluidity
Phospholipids can move laterally;	• fluidity; • can reseal itself after it is disrupted; • can invaginate and form vesicles; • transient gap for movement of small, uncharged &/ non-polar molecules;
cholesterol wedged between phospholipid molecules;	regulates membrane fluidity;
Structure	Property – Cellular identity
• presence of glycolipids and glycoproteins; • carbohydrates chains face the external matrix outside the cell;	• membrane is asymmetrical; • carbohydrate chain serves as cell surface marker / cellular recognition;
;; for 1 mark, 1 mark for every complete description of structure and relating it to the correct property, 6 marks max	

7.

(a) Describe the main features of oxidative phosphorylation.

[7]

- Oxidative phosphorylation comprises ETC & Chemiosmosis;
- NADH and FADH₂ produced in glycolysis, link reaction and Krebs cycle passed their electrons and hydrogen ions to ETC;
- NAD⁺ and FAD are regenerated;
- ETC comprises electron carriers arranged in progressively lower energy level in the inner mit. Membrane
- As electron pass down ETC, energy is released;
- Electron eventually reaches oxygen, which is the final electron acceptor, water is formed;
- Energy is used to pump proton from matrix into intermembrane space and build up a proton gradient;
- Inner mit. membrane is impermeable to protons so proton gradient is maintained;
- Ref to proton motive force
- As protons diffuse down conc. gradient into matrix through ATP synthase found in the inner mit. membrane;
- Energy is released and used by ATP synthase to phosphorylate ADP to form ATP;
- Ref to chemiosmosis;

; for 1 mark, 7 marks max

(b) Describe the main stages of the Calvin cycle.

[6]

- The Calvin cycle incorporates each CO₂ molecule, one at a time, by attaching it to a five-carbon sugar named ribulose biphosphate (RuBP);
- The enzyme that catalyzes this first step is RuBP carboxylase, or rubisco;
- The product of the reaction is a six-carbon intermediate so unstable that it immediately splits in half, forming two molecules of 3-phosphoglycerate;
- Each molecule of 3-phosphoglycerate receives an additional phosphate group from ATP, becoming 1,3-bisphosphoglycerate;
- a pair of electrons donated from NADPH reduces 1,3-bisphosphoglycerate to glyceraldehyde-3-phosphate (G3P);
- Note that for every three molecules of CO₂, there are six molecules of G3P;
- But only one molecule of this three-carbon sugar can be counted as a net gain of carbohydrate, the remaining are used in regeneration of RuBP;
- the carbon skeletons of five molecules of G3P are rearranged by the last steps of the Calvin cycle into three molecules of RuBP;
- spends three more molecules of ATP;
- The RuBP is now prepared to receive CO₂ again, and the cycle continues;

; for 1 mark, 6 marks max

(c) Discuss the social implications of genetically modified organisms.

[7]

(a) Threat to human safety

- Ref. to transgenic food causing allergies with named examples (e.g. Bt toxin in Bt corn/genes in Golden rice coming from several organisms);
- Ref. to long term unexpected/negative effect of transgenic food on human health with named examples;

(b) Threat to safety of the environment

- Ref. to cross-pollination/unintended transfer of transgenes from GM crops/crop-to-weed hybridization with named example (e.g. soy crop plants with herbicide-resistance);
- Ref. to GM crops establishing as weeds/weeds becoming 'superweeds' and thus are invasive/reduces biodiversity/disrupts ecological balance;
- Ref. to GM crops being toxic to non-target organisms with named example (e.g. Bt toxin affecting larvae of monarch butterflies);

(c) Issues pertaining to Access and Intellectual Property

- Ref. to impact of patents (e.g. increase in price of seeds/ domination of world food production by few companies);
- Ref. to biopiracy and how developed countries exploit the resources of developing countries;

(d) Labelling is not mandatory in some countries

- Ref. to need for labelling GM food (e.g. religious/medical/dietary concerns);
- Ref. to varying standards in labelling transgenic food with named examples (e.g. labelling not mandatory in US);

(e) Other social implications

- Ref. to advances being skewed to interests of developed countries/benefiting only rich countries;
- Ref. to dominance of world food production by developed countries / increasing dependence of developing countries on industrialized nations;

; for 1 mark, 7 marks max